



Embedding models

Block 07 · From generative AI back to random effects

Generative AI

Goal: generated data indistinguishable in distribution from real data.

DEMO

Generated people

www.thispersondoesnotexist.com

Every refresh = a new face. None of them exist.

How is that done?

Data is a mix of:

- **Observed quantities** (pixel intensities).
- **Unobserved quantities** (face, smiling, ...).

We humans learn to extract the unobserved.

- GenAI needs to do that too.

How is that done?

Data is a mix of:

- **Observed quantities** y .
- **Unobserved quantities** z — the “latent variables”.

DEFINITIONS

Generative models

DEFINITIONS

- z : latent variables of dimension d .
- y : observed response/data.
- y_g : generated/simulated/synthetic response/data.

MODEL

$$\begin{aligned}y_g &= f(z) + \varepsilon \\z &\sim \mathcal{N}(0, I_{d \times d}) \\\varepsilon_i &\sim \mathcal{N}(0, \sigma^2)\end{aligned}$$

Objective: choose f such that the distribution of y_g is close to the distribution of the observed data y .

DEFINITIONS

Conditional generative models

DEFINITIONS

- z : latent variables of dimension d .
- x : observed covariates.
- y : observed response.
- y_g : generated/simulated/synthetic response.

MODEL

$$y_g = f(z, x) + \varepsilon$$

$$z \sim \mathcal{N}(0, I_{d \times d})$$

$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

Objective: choose f so the conditional distribution of $y_g \mid x$ is close to that of $y \mid x$.

NLME is generative AI!

DEFINITIONS

- η : latent variables of dimension d , covariance Ω .
- x : observed covariates.
- dv : observed response.
- dv_g : generated/simulated/synthetic response.

MODEL

$$dv_g = f_\theta(\eta, x) + \varepsilon$$

$$\eta \sim \mathcal{N}(0, \Omega)$$

$$\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

- A population NLME *is* a conditional generative model.

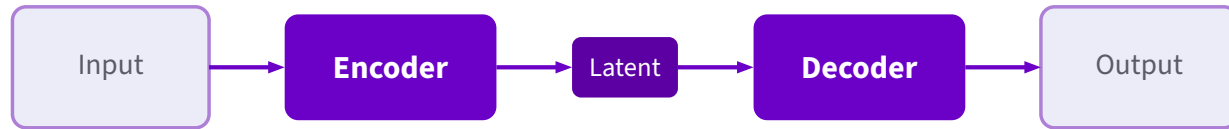
Latent variables = random effects

HOW DO WE FIT?

Maximise the marginal likelihood — marginalise over η .

$$\mathcal{L}(\theta) = \int p(\mathbf{d}v \mid \theta, \eta, x) p(\eta \mid \theta) d\eta$$

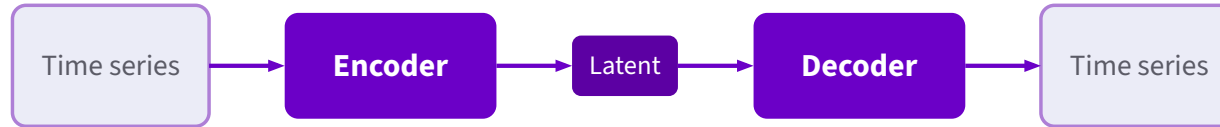
Generative AI — typical anatomy



Input → encoder → latent → decoder → output.

MAPPING

NLME as GenAI



INPUT / OUTPUT

Time series of outcomes / concentrations.

DECODER

The structural NLME — parameter transforms, dynamics, etc.

ENCODER

???

We don't have one. We use marginal-likelihood fitting to back out η instead.

What do “latent variables” represent?

NLME

Interpretation comes from **structural constraints** — e.g. η_1 is a multiplicative deviation on clearance because the model literally writes $CL_i = CL \exp(\eta_{i,1})$.

These interpretations can fool us.

WITHOUT STRUCTURAL CONSTRAINTS

η has no built-in meaning — it is whatever direction in latent space the fit found useful.

What do “latent variables” represent? — image GenAI

For an image, the latent variables are **not** pixel-by-pixel intensity. Rather:

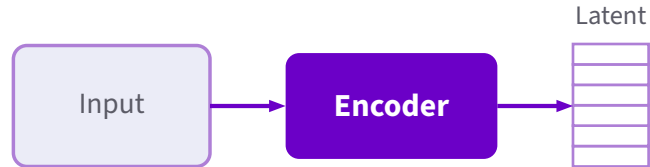
- What objects are in the image.
- What are the characteristics of the objects.
- What are they doing.
- What’s the style.

What do “latent variables” represent? — text GenAI

For text, the latent variables are **not** words. Rather:

- Sentiment.
- Conveyed information.
- Writing style.
- Language.
- ...

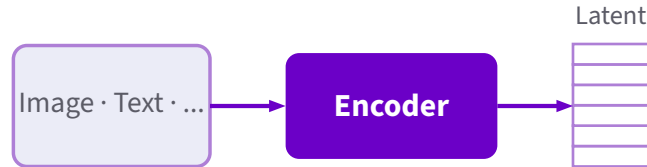
Embedding models — the encoder, alone



Extract meaningful, information-dense **features** from the data — without decoding back to the original modality.

huggingface.co/spaces/mteb/leaderboard

A unified “API” for complex data



Different data modalities (images, text, ...) all map to embeddings — simple vectors of numbers.